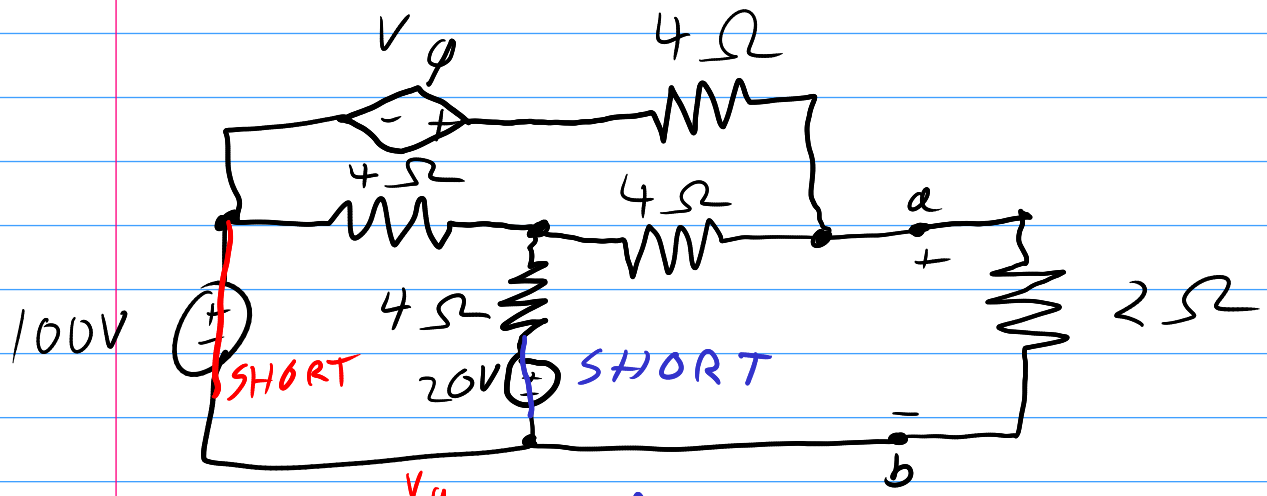
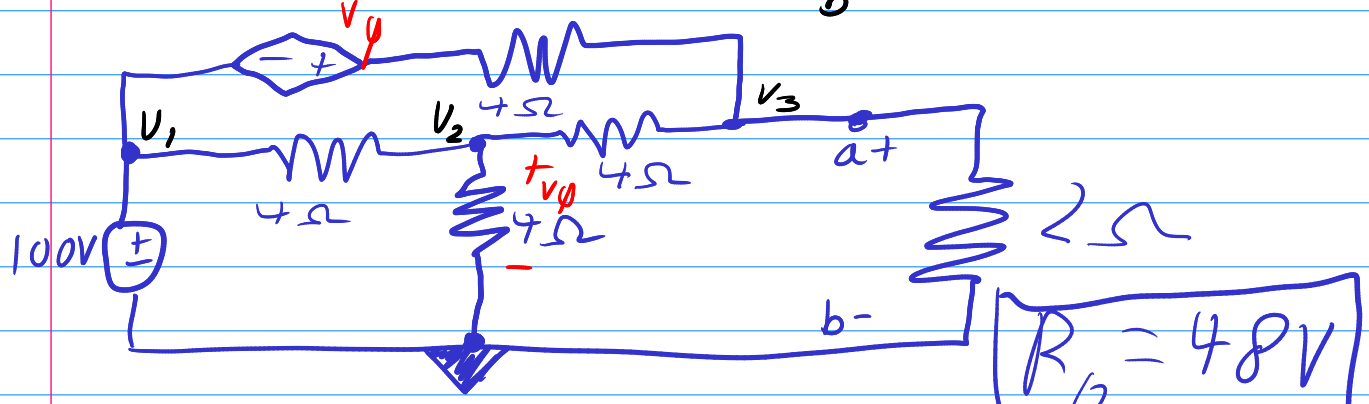




$R_0 = 48V$

$$\frac{V_2-100}{4}+\frac{V_2-0}{4}+\frac{V_2-V_a}{4}=0$$

$$3V_2-100-V_a=0$$

$$V_a=3V_2-100$$

$$\frac{V_a-V_2}{4}+\frac{V_a-(100+V_2)}{4}+\frac{V_a}{2}=0$$

$$4V_a-2V_2-100=0$$

$$2V_a-V_2-50=0$$

$$5V_2=250 \qquad V_2=50V$$

$$V_a=50V$$

$$\frac{V_2-0}{4}+\frac{V_2-20}{4}+\frac{V_2-V_a}{4}=0$$

$$3V_2-20=V_a$$

$$5V_2=30 \qquad V_2=6V$$

$$2V_a-V_2+10=0 \qquad V_a=-2V$$

$$4V_a-2V_2+20=0$$

$$\frac{V_a-V_2}{4}+\frac{V_a-V_2+20}{4}+\frac{V_a}{2}=0$$